Acta Astronautica 200 (2022) 176–187

Contents lists available at ScienceDirect

Acta Astronautica

journal homepage: www.elsevier.com/locate/actaastro

A novel stable and safe model predictive control framework for autonomous rendezvous and docking with a tumbling target

Kaikai Dong [a], Jianjun Luo [a,*], Daniel Limon [b]

[a] Scholl of Astronautics, Northwestern Polytechnical University, Xi'an, 710072, China
[b] Departamento de Ingniería de Sistemas y Automática, Universidad de Sevilla, Sevilla, 41092, Spain

ARTICLE INFO

Keywords:
Safe and stable-by-design
Autonomous rendezvous and docking
Tumbling target
Time variant LOS region
Deterministic convergent
Novel MPC framework

ABSTRACT

This study proposes a novel, safe, and stable-by-design model predictive control (MPC) framework for multistage autonomous rendezvous and docking (AR&D) with a tumbling target, considering several practical challenges (e.g., control saturation, velocity constraints, and collision avoidance) in dock-enabling conditions. In the first stage, global near-optimal and deterministic convergent strategies are designed to drive the chaser to a time-varying line-of-sight (LOS) region within a few steps. The proposed controller includes a terminal constraint to ensure recursive feasibility and stability. In the second stage, the proposed controller is a periodic MPC for tracking under the novel framework, whose reference to be followed is the trajectory of the docking port of the tumbling target. This controller combines trajectory planning and control in a single layer, thereby improving the real-time performance of the algorithm. Moreover, the novel MPC framework incorporates a terminal constraint that guarantees that the closed-loop system enjoys recursive feasibility, safe evolution, and asymptotic convergence to the optimal admissible periodic trajectory with respect to the trajectory of the target docking port. The simulation results verified the efficiency of the proposed control strategy.

1. Introduction

Autonomous rendezvous and docking (AR&D) of two spacecraft is one of the most common scenarios of rendezvous and proximity operations (RPOs), which have been extensively studied in recent years [1,2]. Trajectory planning and control is a critical and important issue for AR&D missions, considering strict engineering requirements, such as velocity constraints, collision avoidance, and control saturation [3–6]. During the AR&D mission, the chaser spacecraft must maintain synchronicity with the target's attitude/pose as it closes its distance to the docking port of the target from the initial position [7–10]. Generally, we can obtain the optimal rendezvous trajectory by solving the optimal control problem [3], considering a multiobjective optimal index, such as the expenditure of fuel and time [11], and/or trajectory robustness [12–14]. Some scholars have presented the artificial potential function as a guidance method to approach the target while obstacle avoidance is fulfilled [15–17]. However, the solution of such a method can easily fall into a local minimum value point. Some scholars applied the indirect method to derive the first-order necessary conditions for optimal control problems using the Pontryagin maximum/minimum principle [18,19]. A fast-guidance method that parameterizes the trajectory and control input, namely the inversion method, was proposed by Ventura and Romano for the AR&D mission [20]. However, these methods cannot simultaneously satisfy the demands of engineering constraints or online implementation [18–20].

The ability to handle constraints systematically and elegantly and use a wide range of prediction models and a solid theoretical framework that ensures closed-loop stability by design makes model predictive control (MPC) a popular method in the academic and engineering practice of spacecraft AR&D field. To the best of our knowledge, Richards et al. were one of the earliest scholars to apply the MPC strategy for controlling spacecraft in rendezvous scenarios [21] and later extended it to a robust case [22]. Hartley presented a novel design and implementation of an MPC system for rendezvous with a passive target, where every phase has its MPC-associate controller [23]. Manni et al. developed an explicit MPC method to improve computational efficiency [24]. A multistep and robust strategy in short-range rendezvous with a statistical target was proposed by Zhu et al. [25], considering the line-of-sight (LOS) constraint.

The problem of AR&D with a non-cooperative target has become increasingly critical, thus attracting increasing attention in recent years. One significant and vital class of non-cooperative targets is space debris,

* Corresponding author.
E-mail addresses: dongkaikai@mail.nwpu.edu.cn (K. Dong), jjluo@nwpu.edu.cn (J. Luo), dlm@us.es (D. Limon).

https://doi.org/10.1016/j.actaastro.2022.08.012
Received 2 March 2022; Received in revised form 24 May 2022; Accepted 6 August 2022
Available online 13 August 2022
0094-5765/© 2022 IAA. Published by Elsevier Ltd. All rights reserved.

such as failed spacecraft, which are of some unknown character or uncontrolled attitude dynamics (e.g., rotating or tumbling), making the problem of AR&D more challenging and complicated. The MPC method was applied to the problem of rendezvous with a rotating platform by some scholars, considering various engineering constraints, such as control saturation, LOS constraints, approaching velocity constraints [26–28], and even chance constraints [29]. Li et al. studied a 3-dimensional space case using the MPC method [30]. A tube-based robust MPC strategy was proposed to rendezvous with a tumbling target in an uncertain environment considering various scenarios [5,31,32].

However, neither of the proposed MPC controllers are stable by design, and terminal constraint satisfaction is not ensured. Moreover, the chaser must maintain its trajectory within a safe LOS cone emitted from the docking port on the target at the final approaching stage of AR&D missions [33,34]. Although it can ensure safety in AR&D missions, the LOS constraint for tumbling targets has not been well studied. The position of the docking port is time-varying; thus, it is a challenge for the controller to drive the chaser to the LOS cone and maintain it to approach the docking port, considering the engineering constraints.

Therefore, this study proposes a novel, safe, stable, and multistage comprehensive trajectory planning and control strategy inspired by the previous MPC for tracking theory [35–37] in the process of AR&D with a non-cooperative target (e.g., a tumbling target). In this study, we considered several time-varying engineering constraints (e.g., LOS constraint), rendezvous accuracy, dock-enabling conditions, and the feasibility and stability of the controller. The main contributions of this study are as follows.

1) The proposed MPC controller/strategy can perform the AR&D mission successfully, ensuring a stable-by-design closed-loop evolution while satisfying constraints and accuracy requirements. The solved trajectory is global near-optimal because the control strategy enables joint optimization for all stages rather than separating it factitiously. Moreover, the system includes a time-variant LOS constraint, which is challenging for controller design and has not been considered in previous studies.
2) The novel MPC controller was designed with a deterministic convergent strategy in the first stage, driving the chaser to the time-varying LOS cone. The controller's feasibility and stability are guaranteed by rendezvous strategies and the addition of terminal constraints, which were not considered previously. Moreover, these properties can be achieved with a small prediction horizon up to the controllability order, making implementing an embedded control system more accessible.
3) In the second stage, the proposed controller is a periodic MPC for tracking, whose reference to be followed is the trajectory of the tumbling target, uniting dynamic trajectory planning and control in a single layer to improve the real-time performance of the algorithm. Moreover, the controller incorporates a terminal constraint guaranteeing that the closed-loop system is stable, recursively feasible, and converges asymptotically to the optimal admissible periodic trajectory with respect to the docking port of the trajectory target.

The remainder of this paper is organized as follows. Section 2 formulates the AR&D mission problem. In Section 3, the proposed multistage rendezvous strategy for tumbling targets is described. Section 4 presents the novel MPC controller design, which contains the first and second stages. Section 5 presents the simulation results and the corresponding analyses/discussions. The main conclusions are presented in Section 6.

2. Problem formulation

The optimal or near-optimal trajectory of an AR&D mission (as shown in Fig. 1) is a feasible solution to the optimal control problem, minimizing the performance index (e.g., the expenditure of fuel and

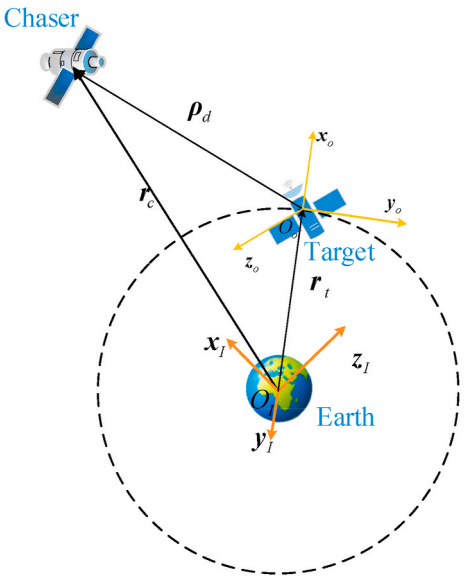


Fig. 1. Schematic of autonomous rendezvous and docking.

rendezvous time), considering some practical engineering constraints. The target considered in this study was an uncontrolled and tumbling object in space (e.g., a malfunctioning satellite/spacecraft) [38]. Information on the tumbling target was assumed to be available and obtained by the chaser's measurement and estimation [39,40]. Readers can refer to the literature in detail regarding the definition of the coordinate frames in the AR&D mission process [5]. The coordinate frames include the Earth-centered inertial coordinate (ECI) frame, local-vertical-local-horizontal (LVLH) frame, target-centered inertial (TCI) frame, and body frame (BF).

2.1. Control objective and relative motion dynamics

In this paper, we study close-range autonomous rendezvous and docking (AR&D) missions in a circular orbit with an orbit altitude less than that of a Geosynchronous orbit (GSO, about 36000 km), and the initial distance between the chaser and the target is of the order of hundreds of meters or smaller. To ensure the success and safety of the AR&D missions, the measurement sensors of the chaser must be maintained pointing toward the docking port of the tumbling target. The objective of the AR&D mission is to drive the chaser to the target docking port. In addition, the chaser must maintain synchronization with the tumbling target in terms of the orientation and angular velocity at the end of the maneuver. Readers can refer to Ref. [30] for detailed docking-enabling conditions. However, the reaction control system can effectively decouple the attitude from the translation [41]. In addition, less fuel is used to synchronize with the target's attitude than to move the chaser during its terminal approach profile [8]. Therefore, the problem of attitude synchronization was not included in this study [8,25,30,41,42].

Assuming that the orbital radius of the target is far greater than the distance between the chaser and target spacecraft, the relative motion of the two objects can be described by the Clohessy–Wiltshire (C–W) equation in the LVLH frame [43]:

$$\begin{aligned} \ddot{x} &= 2n\dot{y} + 3n^2 x + u_x \\ \ddot{y} &= -2n\dot{x} + u_y \\ \ddot{z} &= -n^2 z + u_z \end{aligned} \tag{1}$$

where $\boldsymbol{\rho} = [x \quad y \quad z]^{\mathrm{T}}$ is the relative position and$\dot{\boldsymbol{\rho}} = [\dot{x} \quad \dot{y} \quad \dot{z}]^{\mathrm{T}}$ is the relative velocity. Generally, in engineering practice, if the distance $\|\boldsymbol{\rho}\|_2$ is less than dozens of kilometers, the CW equations are valid.$\mathbf{u} = [d_x, d_y, d_z]^{\mathrm{T}}$ is an external-control acceleration vector.

Eq. (1) is reconstructed in state-space form as follows to facilitate controller design:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} \tag{2}$$

where, $\boldsymbol{x} \in \boldsymbol{R}^6$ is the state, $\mathbf{x} = [\boldsymbol{\rho}^{\mathrm{T}}; \dot{\boldsymbol{\rho}}^{\mathrm{T}}]^{\mathrm{T}}$ and$\mathbf{u} \in \mathbf{R}^3$ is the control vector.

After discretizing the system in Eq. (2) with a sampling period T_s, the discrete-time model is as follows:

$$\mathbf{x}(k+1) = \mathbf{A}_d\mathbf{x}(k) + \mathbf{B}_d\mathbf{u}(k) \tag{3}$$

2.2. Attitude dynamics of the target

Assuming that the tumbling target does not maneuver in orbit. However, there exist attitude maneuvers, since the target is tumbling in space. The body coordinate of the tumbling target is defined as: $O_T x_T y_T z_T$, the origin of which is in the center of the target's mass. The x_T axis, y_T axis, and z_T axis are aligned with its principal body axes. In this study, the tumbling target assumed a rigid, axially symmetric, and slander body that rotates around the principal body axis z_T with a docking port, as shown in Fig. 2. Simultaneously, the z_T axis revolves around the direction of the angular momentum **H**, showing uncontrolled nutation behaviors. The angular momentum **H** of the target remained constant because of the tumbling characteristic in the ECI frame. The principal body axisz_T, angular momentum **H**, and angular velocity $\boldsymbol{\omega}_T$ are in the same plane. The space cone is swept out in inertial space by the angular velocity vector ω_T as it rotates around **H** , whereas the body cone is the trace of $\boldsymbol{\omega}_T$ in the body frame as it rotates about the $\hat{k}_T$ axis. From inertial space, the motion may be visualized as the body cone rolling on the space cone, with the line of contact being the angular velocity vector ω_T [44]. The angle between the axis $\hat{k}_T$ and **H** is the nutation angleθ , while the angle between the axis $\hat{k}_T$ and ω_T is denoted as γ.In Ref. [30], readers can refer to the detailed attitude dynamics of the target.

2.3. Constraints modeling and representation

In engineering practice, all actuators have limited physical control capabilities. The controls are constrained in the magnitude of the three directions and defined as follows [5]:

$$\begin{pmatrix} \mathbf{E}_3 \\ -\mathbf{E}_3 \end{pmatrix}\mathbf{u} \le \mathbf{u}_{\max} \tag{4}$$

where $\mathbf{u}_{\max} = u_{\max}\cdot\mathbf{I}_{6\times1}$ is the maximum magnitude of the control. $\mathbf{I}_{6\times1}$ denotes a 6-dimensional column vector with an element of 1, and $\mathbf{E}_3 \in R^{3\times3}$ is an identity matrix.

In the process of the AR&D mission, we should implement a magnitude constraint on velocity to ensure safe docking. Therefore, the following is the limitation/constraint of velocity:

$$\begin{pmatrix} \mathbf{E}_3 \\ -\mathbf{E}_3 \end{pmatrix}\dot{\boldsymbol{\rho}} \le \mathbf{v}_{\max} \tag{5}$$

where, $\mathbf{v}_{\max} = (\dot{x}_{\max} \quad \dot{y}_{\max} \quad \dot{z}_{\max})^{\mathrm{T}}$ denotes the maximum magnitude of the relative velocity.

Thus, the control saturation and velocity constraints are the time-wise constraints on the states and inputs, which can be written as follows:

$$(\mathbf{x}(k), \mathbf{u}(k)) \in Z \tag{6}$$

where Z is a suitable compact polytope that can easily be derived from (4) and (5). This framework allows us to consider additional constraints that can be posed as a linear combination of the position, velocity, and control actions.

For ultra-close AR&D, the chaser can only approach the docking port of the target from specific directions such as the LOS direction of the target. In this study, an inscribed regular pyramid approximated and linearized the LOS constraint to improve the efficiency of the algorithm [25,29]. Thus, the LOS constraint has a polyhedral formation and is denoted as

$$\Gamma_{los}^{(k)} = \left\{\boldsymbol{\rho} \in R^3 \middle| \mathbf{A}_L(k)\cdot\boldsymbol{\rho} \le \mathbf{b}_L(k)\right\} \tag{7}$$

3. Proposed rendezvous strategy

Because the initial position of the chaser may not be in the LOS cone of the target, the chaser should maneuver to the prescribed safety region $\Gamma_s^{(k)} = \{\boldsymbol{\rho} \in R^3 \middle| \boldsymbol{\rho} \in \Gamma_{los}^{(k)}, \ (x_d, y_d, z_d)\boldsymbol{\rho} \ge \gamma_s\}$ and then approach the docking port within the LOS cone, where (x_d, y_d, z_d) is the position of the docking port in the LVLH frame. The satisfaction of the constraint $(x_d, y_d, z_d)\boldsymbol{\rho} \ge \gamma_s$ implies that $\|\boldsymbol{\rho}\|_2 \ge \gamma_s$, which means the distance from the safety region to the target, should be greater than the distance γ_s. In this study, we propose a reasonable policy to complete the mission based on two stages, the first stage is devoted to steering the chaser to the safe zone, and the second is to drive it to the docking point of the tumbling target, as illustrated in Fig. 3.

The main objective of the controller in the first stage is to steer the chaser to the time-varying safe region $\Gamma_s^{(k)}$. However, this objective is combined with tracking a suitable waypoint $\boldsymbol{\rho}_w(t)$ to achieve better efficiency of trajectory planning in this rendezvous stage, allowing the controller to drive the chaser closer to the target while steering to the safe region, as illustrated in Fig. 4. Therefore, the controller of this stage is posed as a tracking controller that minimizes the tracking error measured by the distance from the chaser to the time-varying safe region and the time-varying waypoint.

Once the chaser has reached an appropriate position inside the LOS region, the second stage can begin. In this stage, the rendezvous strategy is devoted to steering the chaser to the vicinity of the docking port of the tumbling target, while the trajectory of the chaser must be inside the LOS region. In addition, to avoid a possible collision with the target, the trajectory of the chaser must satisfy an additional "keep-out" constraint that ensures that the distance of the chaser to the center of the target is larger than a given radius r_s, along with its evolution. Thereafter, the

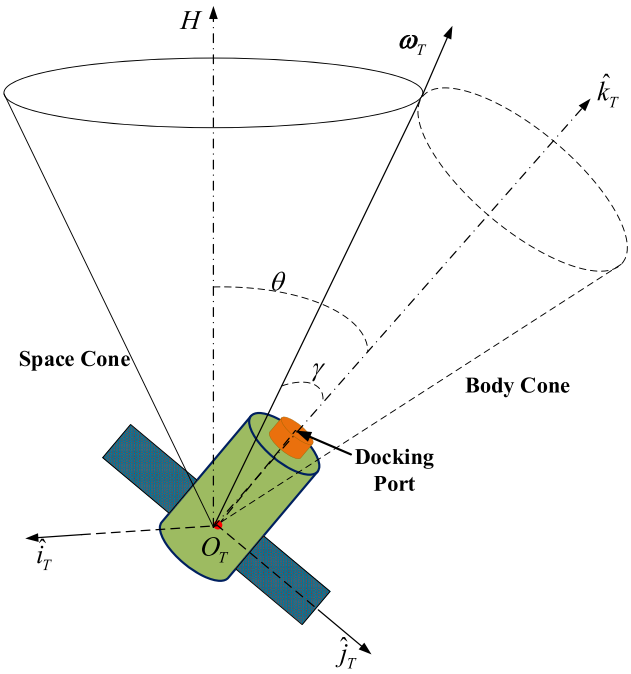


Fig. 2. Tumbling motion of the target.

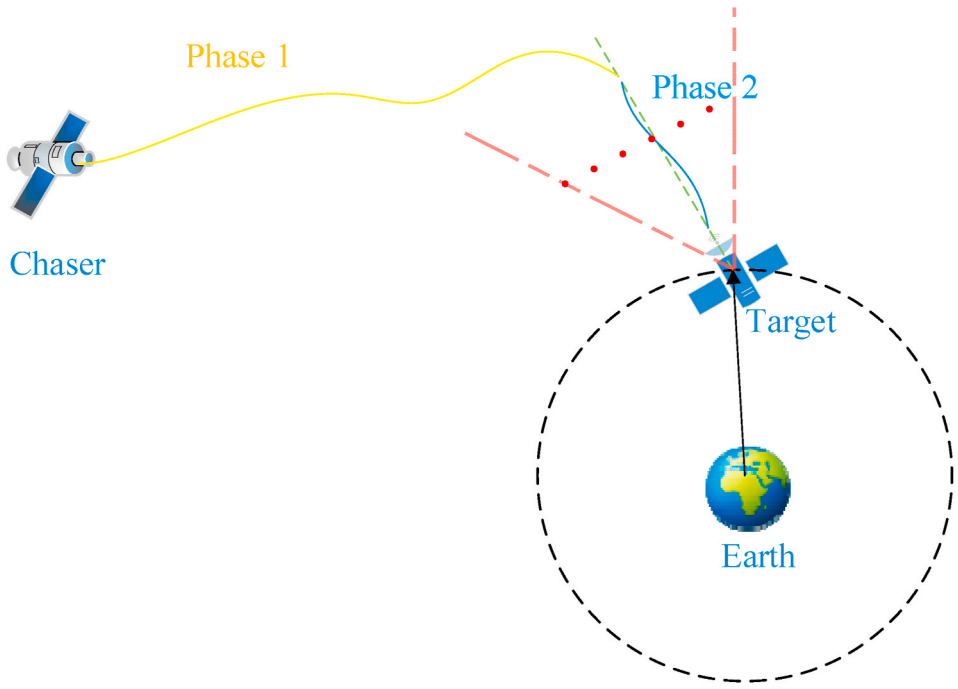


Fig. 3. Simplified geometric representation of the AR&D strategy.

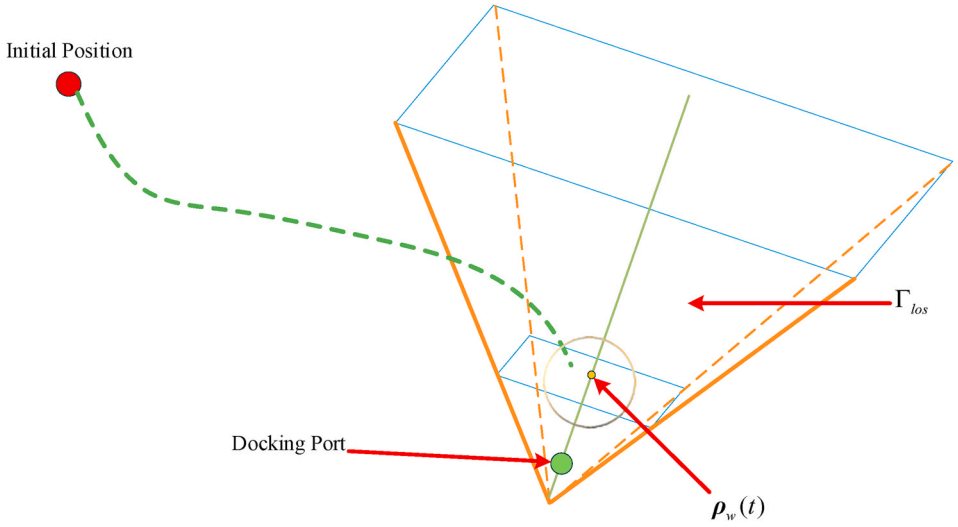


Fig. 4. Illustration of the objective of the first stage.

controller of this stage can be posed as a tracking controller of the periodic trajectory of the docking point of the tumbling target, subject to the hard constraint of maintaining the trajectory inside the region:

$$\Gamma_{kz}^{(k)} = \left\{ \boldsymbol{\rho} \in R^3 \middle| \boldsymbol{\rho} \in \Gamma_{los}^{(k)}, \quad (x_d, y_d, z_d)\boldsymbol{\rho} \geq r_s \right\} \tag{8}$$

The satisfaction of constraint $(x_d, y_d, z_d)\boldsymbol{\rho} \geq r_s$ implies that $\|\boldsymbol{\rho}\|_2 \geq r_s$, which means the chaser should approach the docking point from the outside direction of the "keep-out" zone.

4. Controller design

This section's control policy of the proposed rendezvous strategy for AR&D with the tumbling target is based on MPC for tracking theory. Compared with a standard regulation MPC, MPC for tracking is a novel formulation that includes an artificial steady state and inputs considered as decision variables. This MPC minimizes a cost function that penalizes the deviation between the predicted nominal trajectory and the artificial target, together with the offset cost function that measures the deviation of the artificial target with respect to the target to be tracked. Moreover, a terminal constraint that includes both the terminal state constraint and artificial steady-state and input constraints is added to ensure the stability and feasibility of the controller. The resulting controller can drive the system to any admissible target (either a steady state or a set). If this is not admissible, the state can be steered to the closest possible equilibrium point [35]. The rendezvous strategy and controller design are described in detail below.

In this study, Eq. (3) was used as the prediction model. Thus, given the initial state $x(k)$ of the system and the predicted control sequence $\mathbf{U}(k) = (u^{\mathrm{T}}(k|k), u^{\mathrm{T}}(k+1|k), \ldots, u^{\mathrm{T}}(k+N_p-1|k))^{\mathrm{T}}$ for a specific prediction horizon N_p, the corresponding predicted-state trajectory $\mathbf{X}(k) = (x^{\mathrm{T}}(k+1|k), x^{\mathrm{T}}(k+2|k), \ldots, x^{\mathrm{T}}(k+N_p|k))^{\mathrm{T}}$ can be calculated.

4.1. Controller design of the first stage

The controller is based on MPC for tracking zone regions, where the objective is to steer the system to a given target region Γ_c while minimizing the control effort in the transient [36]. The cost function of the controller includes stage cost function and offset cost function. The stage cost function penalizes the tracking error and the control effort provided by the total fuel expenditure concerning the artificial equilibrium point (x_s, u_s) . The offset cost function penalizes (i) the distance from the artificial output $\boldsymbol{\rho}_s = \mathbf{C}\cdot x_s$ to the target set Γ_c utilizing a convex function $\mathbf{d}(\boldsymbol{\rho}_s, \Gamma_c)$ and (ii) an additional convex term $\mathbf{d}(\boldsymbol{\rho}_s, \boldsymbol{\rho}_w)$ that penalizes the distance from the artificial output to a given waypoint provided by the user. Thereafter, the offset cost function can be denoted as

$$V_o(\boldsymbol{\rho}_s; \Gamma, \boldsymbol{\rho}_w) = \ \widetilde{\beta}_1 \cdot \mathbf{d}(\boldsymbol{\rho}_s, \Gamma_c) + \widetilde{\beta}_2 \cdot \mathbf{d}(\boldsymbol{\rho}_s, \boldsymbol{\rho}_w) \tag{9}$$

where $\widetilde{\beta}_1$ and $\widetilde{\beta}_2$ are tuning constants that allow the user to prioritize convergence to the zone vs. the waypoint. A sensible choice is to set $\widetilde{\beta}_1 \ll \widetilde{\beta}_2$, to get a faster convergence to the waypoint as it approaches the safety zone. $\widetilde{\beta}_2 \cdot \mathbf{d}(\boldsymbol{\rho}_s, \boldsymbol{\rho}_w)$ can be rewritten as $\widetilde{\beta}_2 \cdot \mathbf{d}(\boldsymbol{\rho}_s, \boldsymbol{\rho}_w) = \|\boldsymbol{\rho}_s - \boldsymbol{\rho}_w\|^2_{\mathbf{P}}$, where $\mathbf{P}$ is a positively-defined symmetric matrix.

The resulting cost function of the predictive controller is:

$$V_N(\mathbf{U}_k, x_s, u_s; x_k, \Gamma_c, \boldsymbol{\rho}_w, N) = \sum_{i=0}^{N-1} \left(\|x(k+i|k) - x_s\|^2_{\mathbf{Q}} + \|u(k+i|k) - u_s\|^2_{\mathbf{R}} \right) + V_o(\boldsymbol{\rho}_s; \Gamma_c, \boldsymbol{\rho}_w) \tag{10}$$

where $\mathbf{Q} \in R^{6\times 6}$ and $\mathbf{R} \in R^{3\times 3}$ are the user-prescribed positive-definite weighting matrices. The pair $(x_s, u_s) = \mathbf{M}_\theta \theta$ is the artificial steady state and input, where $\mathbf{M}_\theta$ is the null space of the matrix $[\mathbf{A}_d - \mathbf{I}_6 \quad \mathbf{B}_d]$. In contrast to the classic MPC, $\|x(k+i|k) - x_s\|^2_{\mathbf{Q}}$ and $\|u(k+i|k) - u_s\|^2_{\mathbf{R}}$ in the cost function penalize the deviation from the predicted states and inputs to x_s and u_s, respectively.

Let $P_N^Z(x_k, \Gamma_c, \boldsymbol{\rho}_w, N)$ be the optimization problem, defined as follow:

$$\begin{aligned}
&V_N^*(x_k, \Gamma_c, \boldsymbol{\rho}_w, N) = \min_{\mathbf{U}(k), \theta, \lambda} \sum_{i=0}^{N-1} \left(\|x(k+i|k) - x_s\|^2_{\mathbf{Q}} + \|u(k+i|k) - u_s\|^2_{\mathbf{R}} \right) + V_o(\boldsymbol{\rho}_s; \Gamma_c, \boldsymbol{\rho}_w) \\
&\text{s.t.} \\
&x(k|k) = x_k, \\
&x(k+i+1|k) = \mathbf{A}_d x(k+i|k) + \mathbf{B}_d u(k+i|k) \quad i = 0, \ldots, N-1 \\
&(x(k+i|k), u(k+i|k)) \in Z, \quad i = 1, \ldots, N \\
&(x_s, u_s) \quad = \mathbf{M}_\theta \theta \\
&(x_s, u_s) \in Z \\
&x_{N_p} = x_s
\end{aligned} \tag{11}$$

According to Ref. [30], the optimization problem can be transformed into a quadratic programming (QP) problem. Thereafter, the control law is the first control input of the optimal control sequence $\mathbf{U}^*(k)$ according to the receding horizon policy:

$$\kappa_{\mathbf{MPC1}}(x_k, \Gamma_c, \boldsymbol{\rho}_w, N) = u^*(k|k) \tag{12}$$

To address the time-varying nature of the target, we considered a time-varying target zone and waypoint that is updated at every sampling time according to the policy shown in Algorithm 1. The rationale of this policy is that, initially, the target zone and waypoint are taken as the predicted safe region and waypoint at time $k + N_p$, in such a way that the target is shifted in time. Once the artificial equilibrium point has reached the target zone, it is frozen, and the prediction horizon shrinks at each subsequent sampling time, forcing convergence to the target zone. Readers can refer to Fig. 5 for a graphical description. When the chaser reached the target zone, the controller was ready to switch to the second stage.

Algorithm 1. The control strategy for the first stage of the AR&D mission.

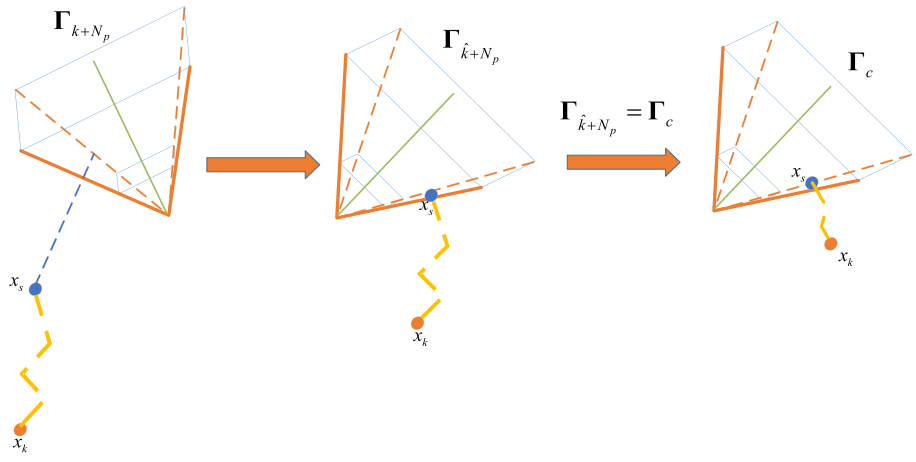


Fig. 5. Rendezvous strategy of the first stage, from left to right, the controller aims to track the time-varying target zone (left figure). Once the terminal state has reached the target zone (center figure), this is frozen, and the state converges to the target zone in N steps by contracting the prediction horizon (right figure).

Initialization: Set $k=0$, $\Gamma_c=\Gamma_s^{k+N_p}$, $\boldsymbol{\rho}_w=\boldsymbol{\rho}_w(k+N_p)$, flag=0, and $N=N_p$.

Step 1. If flag==0:

Then, solve the problem $P_{N_p}^Z(x_k,\Gamma_c,\boldsymbol{\rho}_w,N)$, obtain $x_s^*(k)$ and $\boldsymbol{U}^*(k)$.

Else, set $N=N-1$, solve the problem $P_{N_p}^Z(x_k,\Gamma_c,\boldsymbol{\rho}_w,N)$, obtain $\boldsymbol{U}^*(k)$, and go to **Step** 3.

Step 2. If $\boldsymbol{C}x_s^*(k)\notin\Gamma_c$:

Then, set $\Gamma_c=\Gamma_s^{k+N_p+1}$ and $\boldsymbol{\rho}_w=\boldsymbol{\rho}_w(k+N_p+1)$.

Else, set flag=1.

Step 3. Set $\kappa_{MPC1}(x_k,\Gamma_c,\boldsymbol{\rho}_w,N)=u^*(k\mid k)$, apply the control action, and set $k=k+1$.

Step 4. If $\boldsymbol{C}x_s^*(k)\in\Gamma_c$

Then, go to the **End**.

Else, go to **Step** 1.

End

Remark 1. It is essential to highlight that this algorithm ensures the convergence of the chaser to the target's safe region in a finite number of steps under the assumption that the optimal artificial reference reaches the target region at some point. This assumption is naturally achieved by the design of the controller by virtue of the control authority of a regular spacecraft. From Ref. [45], we have that if the weighting factors ($\widetilde{\beta}_1$ and $\widetilde{\beta}_2$) of the offset cost function V_o are sufficiently large, then the offset cost function plays the role of an exact penalty function that forces the offset cost function of the optimal solution to be null whenever possible. Therefore, the optimal artificial equilibrium point is in the safe region. Once the artificial equilibrium point is in the predicted safe region, it remains constant thereafter. The prediction horizon is shrunk to force convergence to the target region in less than N steps, owing to the properties of the MPC for tracking. This is shown in the illustrative example.

Remark 2. Because the set of constraints of the problem $P_{N_p}^Z(x_k,\Gamma,\boldsymbol{\rho}_w)$ does not depend on the target region Γ and waypoint$\boldsymbol{\rho}_w$, the recursive feasibility of the MPC used in Algorithm 1 is guaranteed irrespective of the time-varying nature of Γ and $\boldsymbol{\rho}_w$.

4.2. Controller design of the second stage

In the second stage, the control object captures the docking point and follows it thereafter while meeting the engineering constraints, particularly the LOS and obstacle avoidance constraints. Because the target is tumbling, the position of the docking port changes with time, as do LOS and obstacle avoidance constraints. We assumed that the trajectory of the target was known a priori and periodic with a known period in the ECI frame. The trajectory of the docking port shows some quasi-periodic character in the LVLH frame because of the slight movement related to the ECI frame. This trajectory of the docking port, denoted as $\mathbf{r}_t$ plays the role of reference to be tracked. Owing to the tumbling characteristic, $\mathbf{r}_t$ may not be reachable, considering the engineering constraints of the chaser.

Therefore, in this stage of the study, we propose a controller based on MPC for tracking periodic signals presented in Ref. [37], which guarantees that the closed-loop system converges asymptotically to the optimal reachable periodic trajectory. The optimal admissible periodic trajectory is defined as the admissible trajectory obtained from the trajectory planner, where a specific criterion is optimized while guaranteeing constraint satisfaction [37]. For a given period N_T, and taking 0 as the initial time of the second stage, the optimal periodic reachable trajectory is defined by its initial state x_0^r and the corresponding sequence of inputs $\mathbf{U}^r=(u^{r\mathrm{T}}(0),u^{r\mathrm{T}}(1),\ldots,u^{r\mathrm{T}}(N_T-1))^{\mathrm{T}}$, and is generally set as:

$$
\begin{aligned}
&\min_{x_0^r,\mathbf{U}^r}\ V_p\left(\mathbf{r}_t;x_0^r,\mathbf{u}^r\right)=\min_{x_0^r,\mathbf{U}^r}\left(\sum_{j=0}^{N_\mathbf{T}-1}\|y^r(j)-\mathbf{r}(j)\|_{\mathbf{S_y}}^2+\sum_{j=0}^{N_\mathbf{T}-2}\|u^r(j+1)-u^r(j)\|_{\mathbf{S_u}}^2\right)\\
&\text{s.t.}\ \ x^r(j+1)=\mathbf{A}_d x^r(j)+\mathbf{B}_d u^r(j)\\
&\quad x^r(0)=x_0^r\\
&\quad y^r(j)=\mathbf{C}x^r(j)\\
&\quad (x^r(j),u^r(j))\in\mathbf{Z}(j)\\
&\quad y^r(j)\in\Gamma_{kz}^{(j)}\\
&\quad x^r(T)=x^r(0)
\end{aligned}
\tag{13}
$$

where, the term $(x^r(j), u^r(j)) \in \mathbf{Z}(j)$ denotes the general engineering constraint, which may be time-varying, and the "keep-out" zone is defined in (12). The cost function measures the deviation between the admissible trajectory and the reference $\|y^r(j) - \mathbf{r}(j)\|_{\mathbf{S}_y}$ and penalizes the increment of the planned reachable input$\|u^r(j+1) - u^r(j)\|_{\mathbf{S}_u}$ to make the planned input trajectory smoother, and thus easier to implement in engineering applications. $\mathbf{S}_y$ and $\mathbf{S}_u$ are symmetric and positively-defined matrices. The optimal admissible trajectory $\mathbf{y}^{r^*}(j)$ is derived from the solution to the optimization problem (13).

The proposed controller for this stage is based on MPC for tracking periodic references theory, which combines the trajectory planner and MPC for tracking in a single optimization problem [37]. Given the receding horizon policy of the predictive controllers, the references provided to the MPC for tracking must be shifted over time. Thus, $\mathbf{r}(k)$ refers to the periodic trajectory of the period N_T to be tracked starting at the current sampling time k and is derived from $\mathbf{r}_t$ by appropriate shifting in such a way that $\mathbf{r}(i) = \mathbf{r}_t(k+i)$.

The optimization problem of the proposed MPC minimizes the cost function $V_{N_p}(x_k, \mathbf{r}(k); x_0^r, \mathbf{U}^r, \mathbf{U}(k))$:

$$V_{N_p}\left(x_k, \mathbf{r}(k); x_0^r, \mathbf{U}^r, \mathbf{U}(k)\right) = V_t\left(x_k; x_0^r, \mathbf{U}^r, \mathbf{U}(k)\right) + V_p\left(\mathbf{r}(k); x_0^r, \mathbf{U}^r\right) \tag{14}$$

where,

$$V_t\left(x_k; x_0^r, \mathbf{U}^r, \mathbf{U}(k)\right) = \sum_{i=0}^{N_p}\left(\|x(k+i|k) - x^r(i)\|_{\mathbf{Q}}^2 + \|u(k+i|k) - u^r(i)\|_{\mathbf{R}}^2\right) \tag{15}$$

$$V_p\left(\mathbf{r}(k); x_0^r, \mathbf{U}^r\right) = \sum_{i=0}^{N_T-1}\|y^r(i) - r(i)\|_{\mathbf{S}_y}^2 + \sum_{i=0}^{N_T-1}\|u^r(i+1) - u^r(i)\|_{\mathbf{S}_u}^2 \tag{16}$$

The term $V_t(x_k; x_0^r, \mathbf{U}^r, \mathbf{U}(k))$ penalizes the tracking error of the open-loop predicted trajectory concerning the artificial reachable reference along the prediction horizon N_p. The term $V_p(\mathbf{r}(k); x_0^r, \mathbf{U}^r)$ penalizes the error between the artificial reachable trajectory and reference signal for one-period N_T. In this application, the prediction horizon N_p was chosen to be lower than the trajectory periodN_T.

To compute the MPC for tracking periodic references $\mathbf{r}(k)$, the following optimization problem is solved at each sampling time:

$$\begin{aligned}
&\min_{x_0^r, \mathbf{U}^r, \mathbf{U}(k)} \quad V_{N_p}\left(x_k, \mathbf{r}(k);\ x_0^r, \mathbf{U}^r, \mathbf{U}(k)\right)\\
&\text{s.t.}\\
&\quad x(k|k) = x_k\\
&\quad x(k+i+1|k) = \mathbf{A}_d x(k+i|k) + \mathbf{B}_d u(k+i|k) \quad i = 0, \ldots, N_p - 1\\
&\quad y(k+i|k) = \mathbf{C}x(k+i|k) \quad i = 0, \ldots, N_p\\
&\quad (x(k+i|k), u(k+i|k)) \in Z \quad i = 0, \ldots, N_p - 1\\
&\quad y(k+i|k) \in \Gamma_{kz}^{(k+i)} \quad i = 0, \ldots, N_p\\
&\quad x^r(0) = x^r\\
&\quad x^r(j+1) = \mathbf{A}_d x^r(j) + \mathbf{B}_d u^r(j) \quad j = 0, \ldots, N_T - 1\\
&\quad y^r(j) = \mathbf{C}x^r(j) \quad j = 0, \ldots, N_T - 1\\
&\quad (x^r(j), u^r(j)) \in Z^c \quad j = 0, \ldots, N_T - 1\\
&\quad y^r(j) \in \Gamma_{kz}^{(j)} \quad j = 0, \ldots, N_T - 1\\
&\quad x^r(0) = \mathbf{A}_d x^r(T-1) + \mathbf{B}_d u^r(T-1)\\
&\quad x\left(k + N_p|k\right) = x^r\left(N_p\right)
\end{aligned} \tag{17}$$

The optimal solution to this optimization problem is denoted as $V_{N_p}^*(x_k, \mathbf{r}(k);\ x_0^{r^*}, \mathbf{U}^{r^*}, \mathbf{U}^*(k))$. The control law is provided by the first input of the optimal reachable predicted trajectory.

$$\kappa_{\mathbf{MPC2}}(x_k, \mathbf{r}(k)) = u^*(k|k) \tag{18}$$

Remark 3. The proposed controller is based on a single layer that unites dynamic trajectory planning and control, and the resulting optimization problem is a QP problem. Thus, the proposed algorithm can be implemented online while making most of the periodical position information of the docking port of the target.

Table 1
Simulation parameters of the target.

Parameters	Value	Unit
Semi-major axis, a	7178160 m	
Eccentricity, e	0.003	
Inclination, i	60	deg
Longitude of the ascending node, Ω	30	deg
Argument of latitude, ϑ	0	deg
Moment of inertia, I_T	diag(100, 100, 60)	kg·m^2
Docking port, r_{doc}	(0,0,1)	m
Initial angles, (ψ_0, φ_0)	(0,0)	deg
Initial rotation rate, $(\omega_{x_0}^{BF}, \omega_{y_0}^{BF}, \omega_{z_0}^{BF})$	(0.27,1.96,1.90)	deg·s^{-1}
The magnitude of angular momentum, $\|H\|_2$	3.14	kg·m^2·s^{-1}
Critical distance, r_s	1	m
Nutation angle, θ	60	deg

Remark 4. The controller is stable by design, and drives the system to converge asymptotically to the optimal admissible periodic trajectory $\mathbf{y}^{r^*}$. In addition, the constraints of the optimization problem solved by the controller do not depend on the reference, allowing for sudden changes in the reference without losing feasibility. Thus, the controller still works when the reference is quasi-periodic.

5. Case study and simulation results

This study designed a multistage, safe, and global near-optimal strategy based on MPC for tracking theory to approach and capture a tumbling target. The initial relative position of the chaser spacecraft (x_0, y_0, z_0) was $(13.08, 30.57, 37.68)$ m in the LVLH frame. Because the distance between the chaser and the target is very small, the initial relative velocity(x_0', y_0', z_0') is $(0, 0, 0)$ m/s. The chaser points to the docking port of the target, rotates around itself, and tumbles initially. The maximum magnitude of the control vector $\mathbf{u}$ was limited to $\mathbf{u}_{\max} = 0.5\text{m/s}^2$ in each direction. The simulation time for the entire AR&D process was 80 s, and the sampling time was 1s. The maximum magnitude of the relative velocity in each direction was set as $v_{\max} = 1$ m/s to ensure the safety of the AR&D mission.

The mission aims to achieve soft, safe rendezvous, and proximity maneuvers with the chaser spacecraft for the tumbling target while

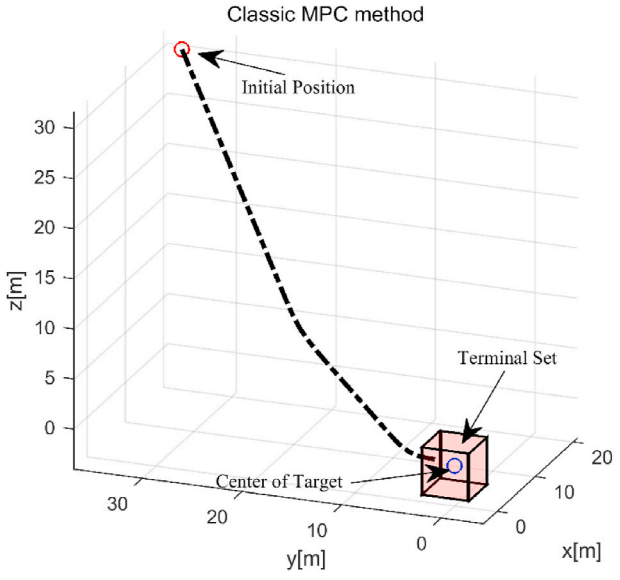


Fig. 6. Trajectory of the chaser spacecraft using the classic MPC method with terminal set.

satisfying various constraints. The simulation parameters for the target are listed in Table 1. All simulations were implemented in MATLAB 2016 and a computer based on Intel(R) CPU i5 @ 2.50 GHz with 8 Gb RAM, using Mosek as the solver, together with the toolbox YALMIP. The computation time was measured by the CPU usage time.

5.1. Classic MPC method

First, this study examined the application of the classic MPC method with a terminal set to the AR&D mission because the design of the terminal set ensured the stability and feasibility of the controller. For simplicity, the target was not considered as tumbling. The LOS constraint was not considered because this test demonstrated the prediction horizon required to accomplish a more straightforward mission for the classic MPC with the terminal set. The simulation parameters were set as described above. The minimum prediction horizon required to make optimization feasible is $N_p = 33$. The closed-loop evolution of the system with prediction N_p is shown in Fig. 6. The red circle represents the initial position of the chaser spacecraft, the blue circle represents the center of the target, and the pink box is the terminal set. The dotted line represents the trajectory of the position of the chaser solved using the classic MPC method.

5.2. Simulation results of proposed controller

Next, we studied the application of the proposed control strategy and method described in Section 3 to an AR&D mission. Detailed simulation results and analyses are described below.

5.2.1. Control strategy of the first stage

In this stage, the objective was to drive the chaser to a prescribed region, Γ_s (part of the LOS cone), whose safe distance to the docking port is greater than γ_s and$\gamma_s = 10$m. The waypoint y_w was set as $y_w = (10, 0, 0)^{\mathrm{T}}$ expressed in the BF of the chaser. The half-cone angle of the LOS constraint was set to 30°. To reduce the computational burden and improve the ability to implement the MPC online, the prediction horizon was set to 6, which is significantly smaller than that required for the regulation MPC. The chaser spacecraft initially points toward the target. The weighting matrices $\mathbf{Q}$ and $\mathbf{R}$ are set as $\mathbf{Q} = \mathbf{I}_6$ and $\mathbf{R} = \mathbf{I}_3$, respectively, while the weighting matrix of the distance $\|\boldsymbol{\rho}_s - \boldsymbol{\rho}_w\|_{\mathbf{P}}$ is chosen as $\mathbf{P} = 236\mathbf{I}_3$.

Fig. 7 shows a snapshot of the trajectories of the chaser, the position of the optimal artificial equilibrium points x_s^*, and the time-varying LOS cone. The red region $\Gamma_{los}^{(k+N)}$ LOS region is the safety target zone in this stage, considering that the current instant is k. Fig. 7 shows that the artificial equilibrium point x_s^* drives toward safety religion $\Gamma_{los}^{(k+N)}$, followed by the chaser. Fig. 8 illustrates the trajectory of the distance $d(x_s^*,$

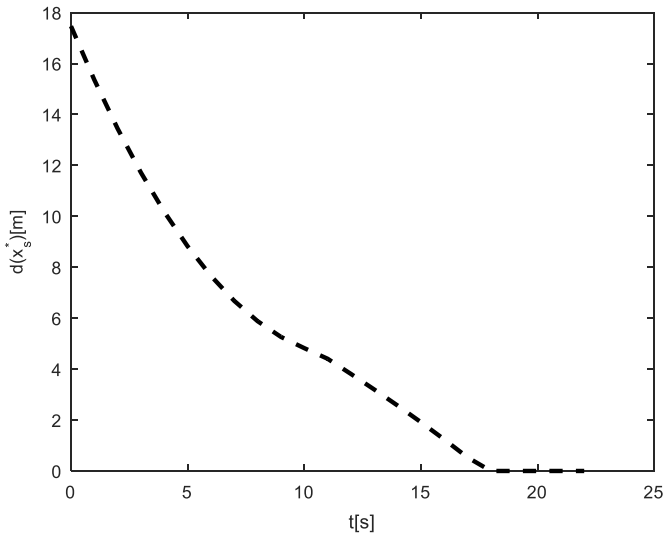


Fig. 8. Trajectory of the distance from the position of the artificial equilibrium point to the safety region.

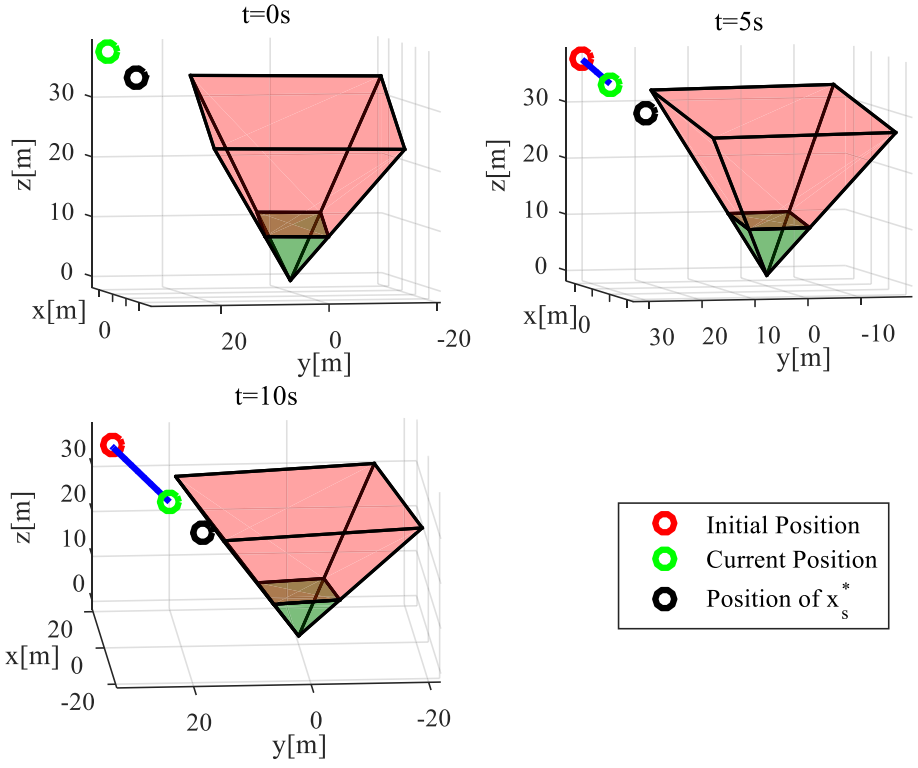


Fig. 7. Snapshot of the trajectories of the chaser and the safety region.

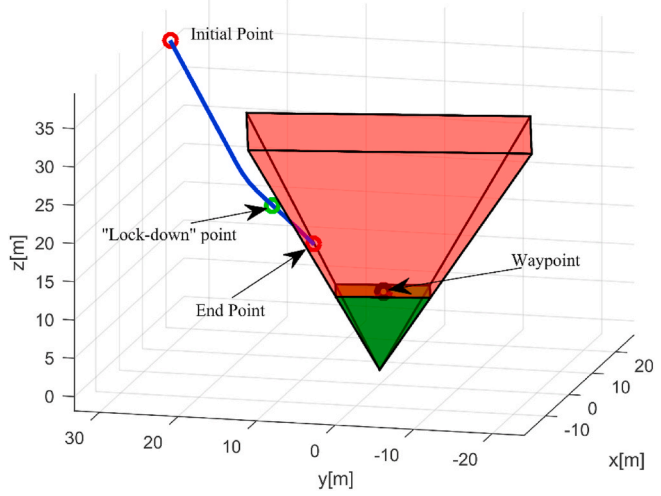


Fig. 9. Trajectory of the chaser spacecraft in the first stage.

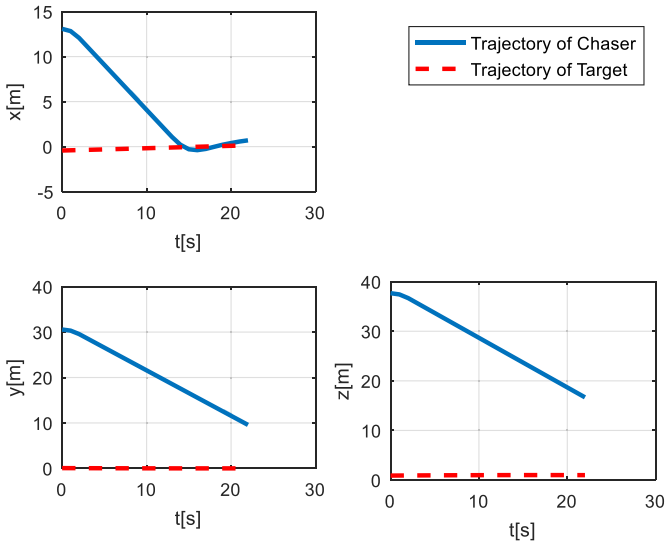


Fig. 10. Trajectories of the relative position of the chaser in the first stage.

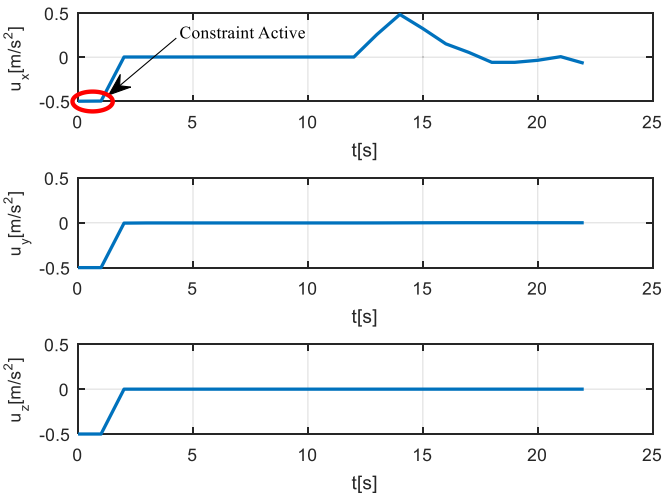


Fig. 11. Thrust profile of the chaser in the first stage.

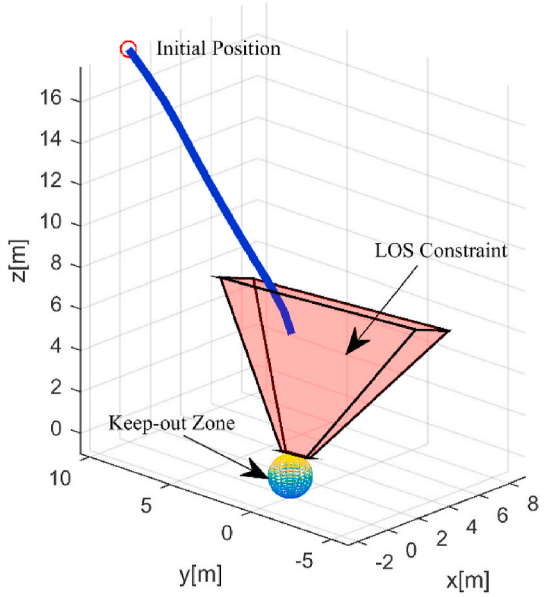


Fig. 12. Relative motion trajectory of the chaser in 32 s in the second stage.

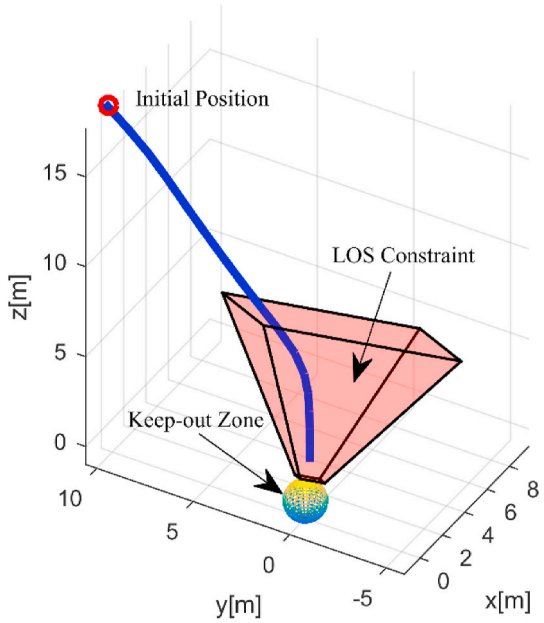


Fig. 13. Relative trajectory of the chaser in 37 s in the second stage.

$\Gamma_{los}^{(k+N)}$) from the position of the artificial equilibrium point x_s^* to the safety region $\Gamma_{los}^{(k+N)}$. From Fig. 8, it can be observed that the distance $d(x_s^*, \Gamma_{los}^{(k+N)})$ decreases as time elapses, and there is an instant at which the distance $d(x_s^*, \Gamma_{los}^{(k+N)})$ becomes 0. Fig. 9 shows the relative motion trajectories of the chaser spacecraft during the first stage of the AR&D mission. The "lock-down" point is when the distance $d(x_s^*, \Gamma_{los}^{(k+N)})$ is 0, considering the time instant k is $\widehat{k}$, depicted by a green point in Fig. 9. Thereafter, the following target safety region $\Gamma_{los}^{(k+N)}$ is frozen $\Gamma_{los}^{(k+N)} = \Gamma_{los}^{(\widehat{k}+N)}$, such that the chaser can go to the time-varying safe region $\Gamma_{los}^{(\widehat{k}+N)}$ in fewer than N steps, as analyzed in the controller design of Section 4. Figs. 8 and 9 show that the chaser "locked" the target region at 18 s and eventually entered the safety region at 23 s, verifying the efficiency and deterministic convergence of the proposed controller/strategy proposed

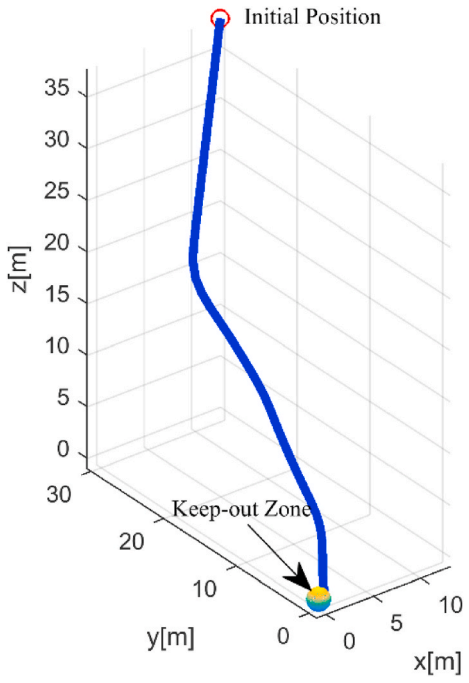


Fig. 14. Relative trajectory of the chaser over the whole stage.

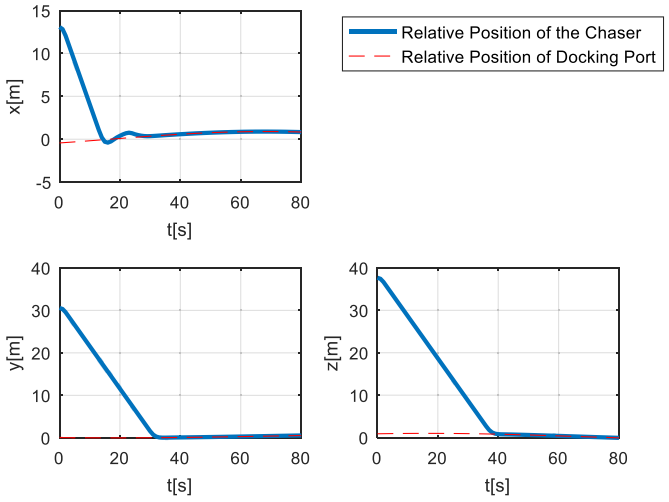


Fig. 15. Trajectories of the relative position of the chaser over the whole stage.

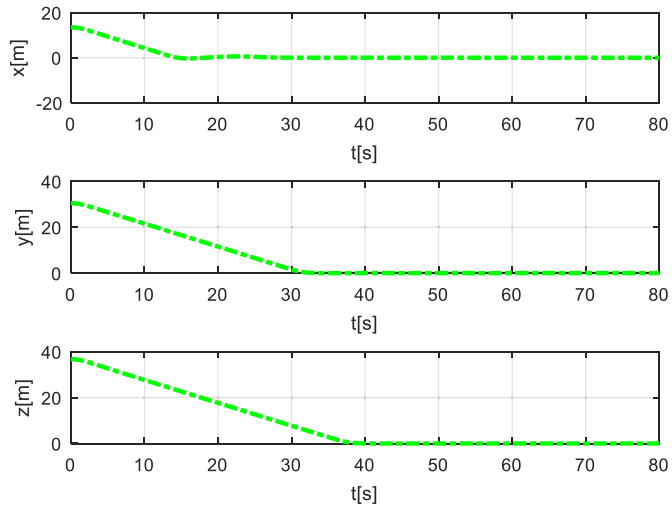


Fig. 16. Tracking errors of the relative position of the chaser over the whole stage.

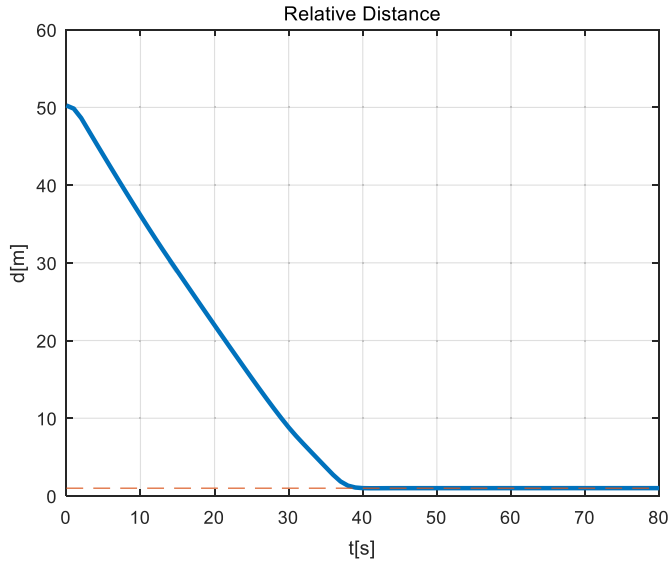


Fig. 17. Tracking errors of the relative position of the chaser over the whole stage.

in this study. Fig. 10 shows the tracking performance of the relative position, whereas Fig. 11 illustrates the thrust profile of the chaser. Fig. 11 shows that the control magnitude satisfies the constraint of control saturation. Figs. 9–11 show that the proposed controller can perform the AR&D mission successfully, considering multiple engineering constraints, particularly time-varying LOS and terminal constraints. The smooth character of the position and control trajectories in Figs. 9–11 and the shorter prediction and control horizon also indicate that the control algorithm is suitable for engineering applications. The average computational cost for the first stage is 0.15 s, which is easy and ideal for online implementation.

5.2.2. *Control strategy of the second stage*

The objective of this stage for the chaser spacecraft is to approach and eventually capture the docking axis of the target along the prescribed direction outside of the "keep-out" zone while meeting the engineering constraints. The weighting matrices $\mathbf{Q}$ and$\mathbf{R}$ in the second stage were set as $\mathbf{Q} = \mathbf{I}_6$ and $\mathbf{R} = \mathbf{I}_3$, respectively. The weighting matrices $\mathbf{S}_u$ and $\mathbf{S}_y$ were designated as $\mathbf{S}_u = \mathbf{S}_y = 1000\mathbf{I}_3$ to drive the planned reachable trajectory smoothly and converge to the reference trajectory quickly. The prediction and control horizons were set to 6. The period length that describes an optimal reachable trajectory in one period was chosen as, $N_T = 200$. In Figs. 12 and 13, the blue lines represent the relative motion trajectories of the chaser spacecraft at 32 s and 37 s, respectively. A ball with a radius of 1 m represents the "keep-out" zone, as explained in Section 2, while the pink region represents the LOS cone.

From Figs. 12 and 13, we can see that the controller can drive the chaser to approach the docking port while meeting the time-varying LOS constraint, which is a hard constraint that can ensure the safety of the AR&D mission. Fig. 14 shows the relative trajectory of the chaser spacecraft. A ball with a radius of 1 m represents the "keep-out" zone, as explained in Section 2. From Fig. 14, we can see that the chaser spacecraft successfully performed the AR&D mission. In addition, the trajectory curve is smooth, which is easy to implement in an actual project, verifying the effectiveness of the proposed algorithm. Figs. 15 and 16

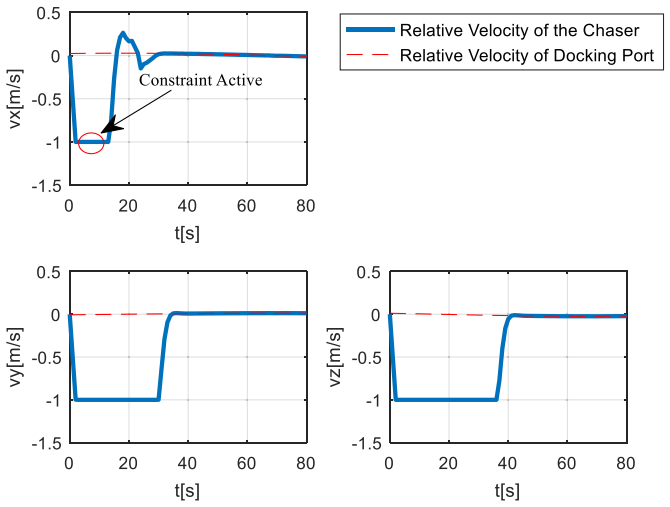


Fig. 18. Trajectories of the relative velocity of the chaser over the whole stage.

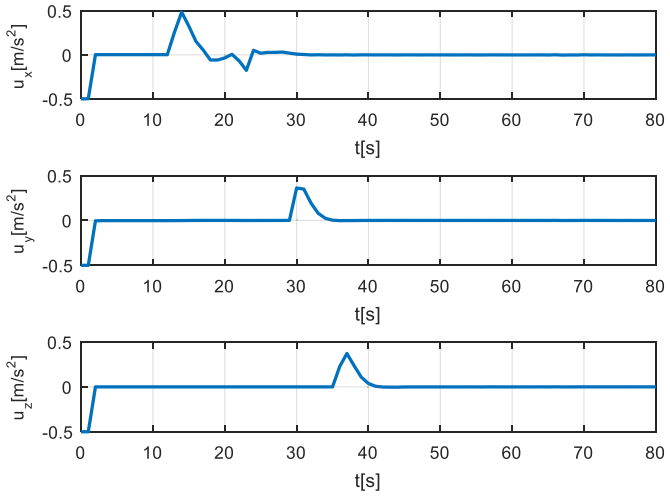


Fig. 19. Thrust profile of the chaser over the whole stage.

represent the tracking performance of the chaser with respect to the relative position and its error, respectively.

The solid-blue line in Fig. 15 is the trajectory of the chaser, whereas the dashed-red line is the trajectory of the docking port of the target. This shows that the chaser gradually approaches the docking port of the target and captures it within 50 s, after which it is synchronized. In addition, the solid-green line and dotted-red line coincide after 40 s, indicating that the tracking accuracy of the proposed algorithm is very high. The green line in Fig. 16 represents the position-tracking error of the chaser. This illustrates that the tracking error gradually converges to zero after 40 s and is maintained thereafter, thereby verifying the effectiveness of the proposed algorithm from another aspect. Fig. 17 depicts the tracking performance of the chaser with respect to the relative distance from the chaser to the center of the target. The solid-blue line is the distance between the chaser and the center of the target, whereas the dotted-red line is that of the docking port. Fig. 17 shows that the position of the chaser gradually converges to that of the docking port. In addition, the chaser approaches the docking port from a prescribed direction outside of the "keep-out" zone, verifying the safety of the proposed algorithm.

Fig. 18 shows the chaser's tracking performance with respect to the chaser's relative velocity. The solid-blue line is the velocity of the chaser and the doted-red line is that of the docking port. From Fig. 18, we can see that the relative velocity of the chaser gradually converges to that of the docking port. Moreover, the trajectory of the relative velocity of the chaser meets the requirement of the approaching velocity constraints because the final approaching velocity should not be very high to ensure the safety of the AR&D mission. Fig. 19 illustrates the thrust profile of the chaser, from it can be observed that the designed controller can satisfy the control saturation constraint, which is usually a hard and physical constraint in engineering. In addition, the smooth curve of the control trajectory indicates that the proposed algorithm is easy to implement in AR&D missions. The average computation cost is 0.30 s for the second stage. Although it requires more time than the first stage, it is still acceptable for online implementation. Moreover, trajectory planning occurs at every sampling time to improve the robustness of the controller, particularly for non-cooperative targets.

6. Conclusion

In this study, a safe and stable MPC strategy/controller is proposed via a multistage comprehensive optimization strategy to accomplish an AR&D mission with a tumbling target in a near-circular orbit. The designed trajectory is globally optimal or near-optimal based on the cost function, considering the chaser's control ability, obstacle avoidance, LOS constraints, etc. The simulation results demonstrate that the proposed control strategy can perform the AR&D mission successfully with a high rendezvous accuracy while considering the engineering constraints, particularly the time-varying and challenging LOS constraints. The control strategy designed in this study is superior to classical MPC methods in the context of a shorter prediction horizon, feasibility, stability, and safety. However, orbital maneuvers of non-cooperative targets and some uncertainties in the process of the AR&D mission are not considered in this study, which is a challenging and meaningful research work, left for future studies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (No.12072269, No. 61690210 and No. 61690211)and the Chinese Scholarship Council. In addition, this study was also supported by grant PID2019-106212RB-C41 found by Ministerio de Ciencia e Innovación /Agencia Estatal de Investigación (MCIN/AEI/10.1309/501100011033), Spain. The authors thank the editor and anonymous reviewers for their constructive comments.

References

[1] P. Lu, X. Liu, Autonomous trajectory planning for rendezvous and proximity operations by conic optimization, J. Guid. Control Dynam. 36 (2) (2013) 375–389, https://doi.org/10.2514/1.58436.
[2] A. Flores-Abad, O. Ma, K. Pham, S. Ulrich, A review of space robotics technologies for on-orbit servicing, Prog. Aero. Sci. 68 (2014) 1–26, https://doi.org/10.1016/j.paerosci.2014.03.002.
[3] R. Chai, A. Savvaris, A. Tsourdos, A review of optimization techniques in spacecraft flight trajectory design, Prog. Aero. Sci. 109 (2019), 100543, https://doi.org/10.1016/j.paerosci.2019.05.003.
[4] P. Li, Z.H. Zhu, Model predictive control for spacecraft rendezvous in elliptical orbit, Acta Astronaut. 146 (2018) 339–348, https://doi.org/10.1016/j.actaastro.2018.03.025.
[5] K. Dong, J. Luo, Z. Dang, Tube-based robust output feedback model predictive control for autonomous rendezvous and docking with a tumbling target, Adv. Space Res. 65 (4) (2020) 1158–1181, https://doi.org/10.1016/j.asr.2019.11.014.
[6] S. Mok, J. Pi, H. Bang, One-step rendezvous guidance for improving observability in bearings-only navigation, Adv. Space Res. 66 (11) (2020) 2689–2702, https://doi.org/10.1016/j.asr.2020.07.035.
[7] M. Leomanni, R. Quartullo, G. Bianchini, Variable-horizon guidance for autonomous rendezvous and docking to a tumbling target, J. Guid. Control Dynam. (2022), https://doi.org/10.2514/1.G006340. Online).

[8] D.C. Sternberg, Optimal Docking to Tumbling Objects with Uncertain Properties, Massachusetts Institute of Technology, Ph.D. Dissertation, 2017.
[9] J. Sanchez, F. Gavilan, R. Vazquez, et al., A flatness-based predictive controller for six-degrees of freedom spacecraft rendezvous, Acta Astronaut. 167 (2020) 391–403, https://doi.org/10.1016/j.actaastro.2019.11.026.
[10] D. Henry, J. Zenteno-Torres, J. Cieslak, et al., A 6-DOF sliding mode fault tolerant control solution for in-orbit autonomous rendezvous, Aero. Sci. Technol. 118 (2021), 107050, https://doi.org/10.1016/j.ast.2021.107050.
[11] X. Chu, Z. Liang, Y. Li, Trajectory optimization for rendezvous with bearing-only tracking, Acta Astronaut. 171 (2020) 311–322, https://doi.org/10.1016/j.actaastro.2020.03.017.
[12] K. Jin, D. Geller, J. Luo, Robust trajectory design for rendezvous and proximity operations with uncertainties, J. Guid. Control Dynam. 43 (4) (2020) 741–753, https://doi.org/10.2514/1.G004121.
[13] R. Volpe, C. Circi, M. Sabatini, GNC architecture for an optimal rendezvous to an uncooperative tumbling target using passive monocular camera, Acta Astronaut. 196 (2022) 380–393, https://doi.org/10.1016/j.actaastro.2020.10.038.
[14] D. Ge, X. Chu, Robust learning for collision-free trajectory in space environment with limited a priori information, Acta Astronaut. 187 (2021) 281–294, https://doi.org/10.1016/j.actaastro.2021.06.038.
[15] D. Zhang, S. Song, R. Pei, Safe guidance for autonomous rendezvous and docking with a non-cooperative target, in: AIAA Guidance, Navigation, and Control Conference, AIAA, 2010, https://doi.org/10.2514/6.2010-7592.
[16] H. Dong, Q. Hu, R. Akella, Safety control for spacecraft autonomous rendezvous and docking under motion constraints, J. Guid. Control Dynam. 40 (7) (2017) 1680–1692, https://doi.org/10.2514/1.G002322.
[17] Y. Guo, D. Zhang, A. Li, Finite-time control for autonomous rendezvous and docking under safe constraint, Aero. Sci. Technol. 109 (2021), 106380, https://doi.org/10.1016/j.ast.2020.106380.
[18] G. Boyarko, O. Yakimenko, M. Romano, Optimal rendezvous trajectories of a controlled spacecraft and a tumbling object, J. Guid. Control Dynam. 34 (4) (2011) 1239–1252, https://doi.org/10.2514/1.47645.
[19] Z. Xu, Y. Chen, Z. Xu, Optimal guidance and collision avoidance for docking with the rotating target spacecraft, Adv. Space Res. 63 (10) (2019) 3223–3234, https://doi.org/10.1016/j.asr.2019.01.026.
[20] J. Ventura, M. Ciarcià, M. Romano, Fast and near-optimal guidance for docking to uncontrolled spacecraft, J. Guid. Control Dynam. 40 (12) (2016) 1–17, https://doi.org/10.2514/1.G001843.
[21] A. Richards, J. How, Performance evaluation of rendezvous using model predictive control, in: AIAA Guidance, Navigation, and Control Conference and Exhibit, AIAA, 2003, https://doi.org/10.2514/6.2003-5507.
[22] A. Richards, J. How, Robust variable horizon model predictive control for vehicle maneuvering, Int. J. Robust Nonlinear Control 16 (2006) 333–351, https://doi.org/10.1002/rnc.1059.
[23] E. Hartley, P. Trodden, A. Richards, Model predictive control system design and implementation for spacecraft rendezvous, Control Eng. Pract. 20 (7) (2012) 695–713, https://doi.org/10.1016/j.conengprac.2012.03.009.
[24] M. Leomanni, E. Rogers, S.B. Gabriel, Explicit model predictive control approach for low-thrust spacecraft proximity operations, J. Guid. Control Dynam. 37 (6) (2014) 1780–1790, https://doi.org/10.2514/1.G000477.
[25] S. Zhu, R. Sun, J. Wang, Robust model predictive control for multi-step short range spacecraft rendezvous, Adv. Space Res. 62 (2018) 111–126, https://doi.org/10.1016/j.asr.2018.03.037.
[26] S. Di Cairano, H. Park, I. Kolmanovsky, Model predictive control approach for guidance of spacecraft rendezvous and proximity maneuvering, Int. J. Robust Nonlinear Control 22 (12) (2012) 1398–1427, https://doi.org/10.1002/rnc.2827.
[27] H. Park, R. Zappulla, C. Zagaris, Nonlinear Model Predictive Control for Spacecraft Rendezvous and Docking with a Rotating Target, 27th AAS/AIAA Spaceflight Mechanics Meeting, 2017.
[28] O.B. Iskender, K.V. Ling, V. Dubanchet, Constraints Tightening Approach towards Model Predictive Control Based Rendezvous and Docking with Uncooperative Targets, 2018 European Control Conference (ECC), IEEE, 2018.
[29] F. Gavilan, R. Vazquez, E.F. Camacho, Chance-constrained model predictive control for spacecraft rendezvous with disturbance estimation, Control Eng. Pract. 20 (2) (2012) 111–122, https://doi.org/10.1016/j.conengprac.2011.09.006.
[30] Q. Li, J. Yuan, B. Zhang, C. Gao, Model predictive control for autonomous rendezvous and docking with a tumbling target, Aero. Sci. Technol. 69 (2017) 700–711, https://doi.org/10.1016/j.ast.2017.07.022.
[31] M. Mammarella, E. Capello, H. Park, Tube-based robust model predictive control for spacecraft proximity operations in the presence of persistent disturbance, Aero. Sci. Technol. 77 (2018) 585–594, https://doi.org/10.1016/j.ast.2018.04.009.
[32] C. Buckner, R. Lampariello, Tube-based model predictive control for the approach maneuver of a spacecraft to a free-tumbling target satellite, in: 2018 Annual American Control Conference (ACC), IEEE, 2018, https://doi.org/10.23919/ACC.2018.8431558.
[33] A. Weiss, M. Baldwin, R. Erwin, Model predictive control for spacecraft rendezvous and docking: strategies for handling constraints and case studies, IEEE Trans. Control Syst. Technol. 23 (4) (2015) 1638–1647, https://doi.org/10.1109/TCST.2014.2379639.
[34] J. Sanchez, F. Gavilan, R. Vazquez, Chance-constrained model predictive control for near rectilinear halo orbit spacecraft rendezvous, Aero. Sci. Technol. 100 (2020), 105827, https://doi.org/10.1016/j.ast.2020.105827.
[35] D. Limon, I. Alvarado, T. Alamo, MPC for tracking piecewise constant references for constrained linear systems, Automatica 44 (9) (2008) 2382–2387, https://doi.org/10.1016/j.automatica.2008.01.023.
[36] A. Ferramosca, D. Limon, A.H. González, MPC for tracking zone regions, J. Process Control 20 (4) (2010) 506–516, https://doi.org/10.1016/j.jprocont.2010.02.005.
[37] D. Limon, M. Pereira, D.M. de la Pena, MPC for tracking periodic references, IEEE Trans. Automat. Control 61 (4) (2016) 1123–1128, https://doi.org/10.1109/TAC.2015.2461811.
[38] L. Felicetti, P. Gasbarri, A. Pisculli, Design of robotic manipulators for orbit removal of spent launchers' stages, Acta Astronaut. 119 (2016) 118–130, https://doi.org/10.1016/j.actaastro.2015.11.012.
[39] Q. Feng, Z. Zhu, Q. Pan, Pose and motion estimation of unknown tumbling spacecraft using stereoscopic vision, Adv. Space Res. 622 (2018) 359–369, https://doi.org/10.1016/j.asr.2018.04.034.
[40] L. Sun, Z. Zheng, Adaptive relative pose control for autonomous spacecraft rendezvous and proximity operations with thrust misalignment and model uncertainties, Adv. Space Res. 59 (7) (2017) 1861–1871, https://doi.org/10.1016/j.asr.2017.01.005.
[41] A. Caubet, J. Biggs, An Inverse Dynamics Approach to the Guidance of Spacecraft in Close Proximity of Tumbling Debris, 66th International Astronautical Congress, International Astronautical Federation. Paper ID, 3116, 2015.
[42] X. Chu, J. Zhang, S. Lu, Optimized collision avoidance for an ultra-close rendezvous with a failed satellite based on the Gauss pseudospectral method, Acta Astronaut. 128 (2016) 363–376, https://doi.org/10.1016/j.actaastro.2016.07.011.
[43] W.H. Clohessy, R.S. Wiltshire, Terminal guidance system for satellite rendezvous, J. Aero. Sci. 27 (1960) 653–674, https://doi.org/10.2514/8.8704.
[44] H. Curtis, in: Orbital Mechanics for Engineering Students, Elsevier Butterworth-Heinemann, Burlington, 2005.
[45] D. Limon, A. Ferramosca, I. Alvarado, Nonlinear MPC for tracking piece-wise constant reference signals, IEEE Trans. Automat. Control 63 (11) (2018) 3735–3750, https://doi.org/10.1109/TAC.2018.2798803.